1. Description of the Project

The Health Misinformation Detector (HMD) operates as a machine learning system which evaluates the trustworthiness of health information found online. Users face difficulties when trying to identify genuine medical information from false or made up content that spreads quickly through social media and web platforms. The system uses automated processing to evaluate health statements for credibility through text analysis and feature extraction and prediction algorithms. The system uses supervised learning models with modern NLP techniques to generate easy to understand predictions about statement credibility.

The project implements a complete workflow by uniting data preprocessing with feature engineering and model development. Users can access the system through a user friendly interface which accepts text entries for analysis by the trained classifier that uses multiple labeled datasets. The system shows how computational methods can fight health misinformation while enhancing web content quality and enabling people to make better decisions.

2. Significance of the Project

Health misinformation creates a major problem in modern digital society because false medical information spreads quickly through social media platforms and online forums and news websites. Health related misinformation affects actual behaviors because people base their vaccine choices and treatment decisions and daily health routines on false information. The spread of false information results in avoidable diseases and postponed medical services and widespread public misunderstandings. The protection of people and their ability to make educated decisions requires the development of methods which identify trustworthy health information.

The research holds importance because it supports current initiatives to build confidence in reliable medical information. The research focuses on credibility assessment to support digital health literacy improvement and user friendly navigation through conflicting health information. The ability to evaluate credibility directly affects health results and communication effectiveness and strengthens the stability of public health systems when people encounter deceptive yet convincing medical information.

3. Code Structure

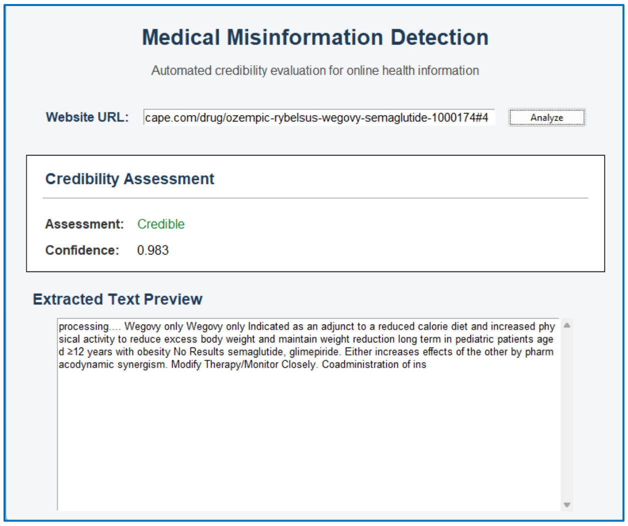
The HMD repository contains three main directories which include backend and frontend and datasets with configuration files located at the project root. The backend directory contains the main processing functions which handle user URL submission and article text extraction and analysis preparation and credibility prediction through model training. The backend system controls the entire process from user URL entry to prediction output by executing tasks including web scraping and data cleaning and classification operations.

The frontend directory enables users to enter URLs and display credibility assessment results through its user friendly interface. The system enables users to assess health content online through a simple interface which does not require knowledge of the internal model operations. The datasets directory contains health data which serves for model training and assessment purposes through its collection of labeled examples between authentic and deceptive content. The requirements.txt file located at the root directory specifies project dependencies while the README.md document provides instructions for setting up and using the project. The system maintains a organized design which divides functions into three distinct sections that handle user interaction and data processing and storage operations.

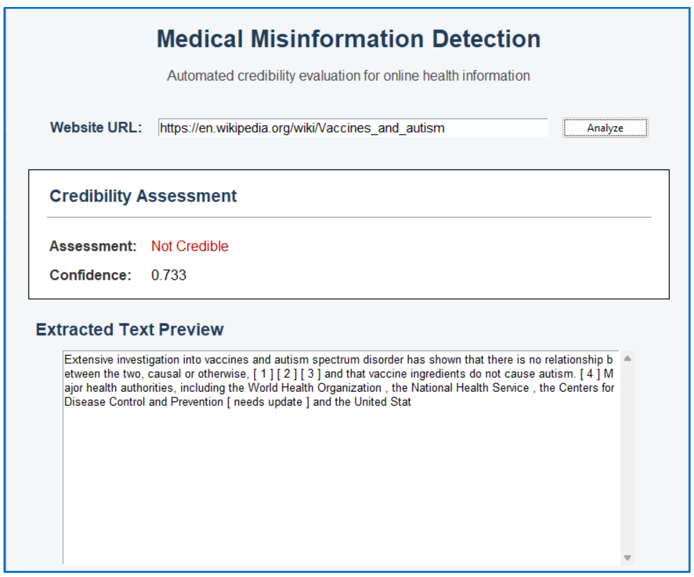
4. Functionalities and Test Results

The program's frontend lets a user detect medical misinformation by submitting a URL link to a website or article. The program displays an error message if the information provided from the URL is either insufficient in length or material. URLs that are accepted are then assessed using our model. They are given a more tangible assessment, through displaying either "Credible" or "Not Credible" depending on the output of the analysis. A confidence score is also displayed, which addresses the certainty of the output. The confidence score is meant to help users in making more informed decisions rather than relying solely on the label output from the program. Our GUI also features an "Extracted Text Preview" box that displays a small section of the scraped text, to ensure that proper information was extracted. The interface was designed to be simple to use, but still professional. The GUI also supports multiple consecutive URL checks without restarting the program, which makes the use of it much easier for researchers that need to assess several sources.

Credible Case



Non-credible Case



5. Data Collection

The Health Misinformation Detector used various domain specific datasets to build a diverse collection of authentic and false health information. The project used FakeHealth and SciFact and KINIT and Credible as its main datasets to create a training set that included both scientifically proven content and actual misinformation examples. The training set includes both scientifically backed content and real world examples of misinformation through the different health information quality perspectives from each source. The FakeHealth dataset contains thousands of health related articles which receive credibility labels to provide extensive collection of untrustworthy health content from online platforms. SciFact delivers professionally authenticated scientific statements together with supporting evidence statements which create high quality factual content that experts have validated. The KINIT dataset provides extra credibility information about public health related claims and articles through its repository which includes both raw data and preprocessed data for model consumption.

The dataset received its main content through manual web scraping operations in addition to using external resources. The Credible dataset contains medical articles which researchers obtained directly from reputable medical institutions including CDC and Johns Hopkins Medicine and Mayo Clinic and NIH and WHO and Yale Medicine. The sources were selected because they publish medical information that undergoes strict review and evidence based evaluation. The system stored each scraped article with title information and URL and publication origin data to provide both transparency and tracking capabilities.

The final collection contains thousands of labeled health information samples which evenly distribute between authentic and false content. The repository contains all files which follow a structured directory structure to support both understanding and duplicate results. The training data includes information from multiple sources which enables the model to learn diverse linguistic patterns and claim types and credibility indicators for effective misinformation detection.

6. Data Processing and Feature Engineering

The machine learning model required multiple data sources to undergo extensive preprocessing for achieving dataset consistency and text preparation. The preprocessing system started by performing text normalization followed by HTML removal and non-informative element extraction from navigation menus and duplicated boilerplate content and formatting symbols. The system converted all content to lowercase for uniformity while removing URLs and non-alphabetic sequences and stray punctuation that did not affect semantic value. The system removed stop words to decrease noise levels while it discarded documents containing missing text or unworkable formatting. The preprocessing step combined all text data from claims and articles and metadata into standardized JSON or CSV formats which created a single data structure for all datasets regardless of their origin.

The feature engineering process converted unprocessed text data into numerical formats which the model could learn from. The model used transformer-based encoding to process each cleaned document and claim which enabled it to understand word relationships and contextual understanding instead of using basic word frequency analysis. The model received special tokens which served to identify document segments while preserving structural information during the encoding process. The model used dense text embeddings as its main representation but it also analyzed scientific terminology and linguistic patterns that indicate misinformation when relevant data was available. The processing system transformed all datasets including institutional articles and SciFact claims and FakeHealth samples and KINIT data into an equivalent feature space that contained valuable information for classification tasks.

7. Model Development

The model we built is meant to be capable of detecting misinformation from medicine and health related text, classifying it as either credible or not credible. Prior to building the model, we collected and preprocessed multiple datasets, including SciFact, KINIT, FakeHealth, and other sources. Each dataset was cleaned to remove noise like HTML tags and scripts. We also split longer articles into smaller chunks to make them easier for the model to process.

For deep learning, we used the DistilBERT language model. DistilBERT is a faster version of the BERT model, which felt like a more appropriate model for our project. DistilBERT uses six transformer layers, while BERT uses 12. The text scraped from the websites was first tokenized using DistilBERT's tokenizer, which converted words into a format the model could understand. Each input was padded and truncated to a maximum length of 256 tokens so that input sizes were uniform. To handle class imbalance, we calculated class weights and used a custom training loop that applied these weights to the loss function. Many datasets had more credible than non-credible examples, or vice versa. Class weights were computed from the training data and applied in the cross-entropy loss, which gave higher penalty to misclassified samples. We trained the model for multiple epochs, and checked metrics like accuracy, precision, recall, and F1 score to check that the model was learning effectively.

A character-level logistic regression model was also used in our project. This model used a TD-IDF vectorizer to convert sequences of 3-6 characters n-grams into numerical features. Logistic regression was then training on these features with balanced class weights to handle any imbalances in the dataset.

8. Discussion and Conclusions

In doing our project, we found that credibility classification for health-related documents and websites was able to be done using both deep learning and traditional machine learning. DistilBERT, which was our deep learning model, was able to detect contextual and semantic information from text, which was important in differentiating credible and non-credible writing. The use of class weights during training helped fix the issue of imbalanced datasets, which improved the performance of the minority class. The character-level logistic regression model also provided a simpler baseline that still performed well, despite being less powerful than DistilBERT. Merging datasets from different sources was important in making sure that the model was exposed to a wide variety of writing styles and topics.

The models developed in our project can serve as a baseline for other applications, like content moderation and research analysis. Future work on this project could include using multiple models in our final output. We could also include graphics and descriptions of documents within our GUI.

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